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PROJECT WORK PHASE - 1 (BIS685) REPORT ON

"Advantage OS: An AI-Driven Local Growth Platform for SMEs"

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CERTIFICATE

This is to certify that Project Work Phase - 1 (**BIS685**) Report entitled “**Advantage OS: An AI-Driven Local Growth Platform for SMEs**” is a bonafide work carried out by **Dhanush H S [1JS23IS036]**, **Gagan G Pawar [1JS23IS046]**, **Harsha L [1JS23IS053]**, **Jhenkar G R[1JS23IS060]**, in partial fulfillment for the award of the degree of Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University Belagavi during the year 2025- 2026.

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ABSTRACT

Small and Medium Enterprises (SMEs) such as cafés, boutiques, and salons face significant challenges in keeping pace with rapidly evolving local consumer trends. Unlike large enterprises, SMEs typically lack access to integrated analytics platforms, campaign intelligence tools, and inventory forecasting systems. Advantage OS is an AI-driven Local Growth Platform designed as a unified Growth-as-a-Service solution that bridges the gap between social media trend discovery and offline retail execution.

The platform collects and processes public social media data from Reddit, YouTube, and other streams, applies Natural Language Processing (NLP) techniques including Latent Dirichlet Allocation (LDA) topic modeling and LSTM-based time-series forecasting to identify hyper-local trends and predict their lifecycle. An Influencer Matchmaker module uses K-Means clustering and bot detection algorithms to surface micro-influencers suited to each trend. A Creative Studio enables AI-generated advertising content using Stable Diffusion/DALL-E for images and GPT-style models for captions, with TensorFlow.js providing in-browser CTR estimation. An Inventory Edge module leverages OCR (Tesseract.js) on receipt images to maintain near-real-time stock records and triggers automated WhatsApp alerts via the Twilio API when stockout risk is detected.

The system is implemented as a modular full-stack application with a React + Tailwind CSS frontend, a Python FastAPI backend, and a Supabase (PostgreSQL) database. It operates as a closed-loop system where social trend signals flow through campaign creation and influencer outreach and connect back to inventory readiness and alert generation. The expected outcome is a functional prototype that demonstrates how SMEs can identify actionable trends earlier, launch aligned campaigns faster, and avoid trend-driven stockouts — all from a single, non-technical-friendly dashboard.

Keywords: SME Marketing, Hyper-local Trend Detection, LDA Topic Modeling, LSTM Forecasting, Influencer Matching, AI Content Generation, OCR Inventory Tracking, WhatsApp Alerts, Closed-loop System, Growth-as-a-Service

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Chapter 1

INTRODUCTION

1.1 Background

The digital economy has fundamentally transformed how consumers discover products and services, yet Small and Medium Enterprises (SMEs) remain disproportionately underserved by the tools available to navigate this transformation. While large corporations invest heavily in dedicated analytics teams, enterprise-grade CRM platforms, and AI-driven marketing suites, local businesses such as neighbourhood cafés, independent boutiques, and salon chains continue to rely on manual observations, word-of-mouth intelligence, and generic social media scheduling tools.

The rise of hyper-local digital communities on platforms like Reddit, YouTube, and Instagram has created a new opportunity: neighbourhood-level conversations now carry measurable signals about what products, aesthetics, or experiences are gaining traction in specific localities. A viral coffee trend on local Reddit sub-groups, for instance, can translate directly into increased footfall for a café that identifies and acts on it quickly. Similarly, a fashion micro-trend surfacing in regional YouTube content can pre-signal demand spikes for boutiques weeks before mainstream media coverage.

Despite this opportunity, the gap between social listening and operational execution remains wide. SMEs typically lack the technical capabilities and the integrated tooling needed to close this loop — from detecting a trend, to generating a campaign, to managing the inventory that supports it. This project, Advantage OS, is proposed to address exactly this gap.

1.2 Overview of Advantage OS

Advantage OS is an AI-Driven Local Growth Platform for SMEs. It is designed as a unified Growth-as-a-Service solution that converts local social media activity into practical marketing and inventory decisions. The platform integrates four major functional modules: a Trend Engine, an Influencer Matchmaker, a Creative Studio, and an Inventory Edge, all connected through a closed-loop feedback architecture.

The Trend Engine collects data from public social media APIs (Reddit via PRAW and YouTube Data API), preprocesses it, applies LDA topic modeling to surface emerging topics, and forecasts trend

lifecycles using LSTM time-series models. This gives SME owners a scored, prioritised view of which trends are actionable in their locality right now.

The Influencer Matchmaker clusters micro-influencer profiles by niche, engagement rate, and locality using K-Means clustering, and applies rule-based bot detection to filter out low-quality accounts. The Creative Studio allows SME users to generate on-brand ad images via Stable Diffusion or DALL-E APIs and AI-generated captions via GPT-style models, with browser-side CTR estimation powered by TensorFlow.js. The Inventory Edge digitises stock management through OCR-based receipt scanning (Tesseract.js) and fires proactive WhatsApp alerts via the Twilio API when a trend-driven stockout risk is detected.

1.3 Context and Relevance

India's SME sector, comprising over 63 million enterprises and contributing approximately 30% of GDP, is one of the most significant yet digitally underserved segments of the economy. Bengaluru, the chosen pilot city for this project, has a vibrant SME ecosystem across F&B, fashion, and wellness — sectors where trend responsiveness directly correlates with business performance.

The relevance of Advantage OS is multifold. From a business perspective, it targets a real, unmet need: an affordable, non-technical-friendly platform that gives local shops the same trend intelligence and campaign velocity that large brands enjoy. From a technical perspective, it presents an academically novel architecture that combines NLP, time-series forecasting, computer vision, generative AI, and OCR within a single operational loop — a combination not seen in existing SME-oriented tools.

The project also aligns with the broader academic conversation around AI democratisation — the idea that the benefits of advanced machine learning should not be concentrated in large enterprises but should be accessible to small business owners who may have little to no technical background.

1.4 Scope of the Study

The current phase of the project focuses on designing and validating the architecture, defining modules and data flows, establishing requirement specifications, and conducting a literature review that contextualises the work within existing research. The scope covers social trend data ingestion from

Reddit and YouTube, trend detection and forecasting, influencer discovery, AI-assisted creative generation, OCR-based inventory tracking, and WhatsApp alert delivery.

The system is scoped as a prototype-level application targeting Bengaluru-based SMEs in the food and beverage, fashion, and wellness sectors. Advanced production-grade features such as multi-tenant billing, dynamic pricing engines, geo-local heatmaps at city scale, and supplier management are explicitly out of scope for this phase.

1.5 Organisation of the Report

This report is organised as follows: Chapter 2 presents the literature survey that informed the design of Advantage OS. Chapter 3 identifies the problem statement in detail. Chapter 4 lists the project objectives. Chapter 5 describes the methodology including system architecture, module design, and development approach. Chapter 6 specifies the software and hardware requirements. Chapter 7 concludes the report with a summary and directions for Phase 2.

Chapter 2

LITERATURE SURVEY

A review of the available references and project materials shows that the proposed work stands at the intersection of retail forecasting, social media trend analysis, recommender systems, and multimodal advertising intelligence. The presentation cites recent studies on demand forecasting, predictive sales analysis, social media trend prediction, topic modeling, and AI-assisted advertising, which together provide the conceptual base for the project architecture.

2.1 Paper 1 : Agentic AI-Driven Demand Forecasting in Retail Systems

Results : Agentic AI and predictive analytics significantly improve retail demand forecasting accuracy, reducing forecast error, stockouts, and enabling better inventory and pricing decisions for large retail systems.

Limitations : The work is focused on back-end forecasting for larger or enterprise retailers and does not integrate social media trend discovery, influencer analytics, or creative campaign workflows needed to close the loop for SMEs.

2.2 Paper 2 : AI in Social Media Marketing Using Machine Learning to Analyse Trends and Predict Consumer Sentiment

Results : ML-based analysis of social media features can predict consumer sentiment and social trends, helping brands understand audience reactions and optimise social media content strategy.

Limitations : It remains limited to marketing insight and content optimisation, without linking trend and sentiment outputs to inventory management or providing an end-to-end workflow from social listening to store-level operational decisions.

2.3 Paper 3 : AI Integrated Recommender System for Improving E-commerce Experience

Results : AI-based recommender systems personalise product suggestions based on user behaviour and preferences, improving customer experience, engagement, and sales in e-commerce platforms.

Limitations : The solution is confined to online recommendation quality and personalisation, ignoring hyper-local social trends, micro-influencer selection, and physical store inventory constraints that matter for SME retail.

2.4 Paper 4 : InvadeAI – Interactive Advertising with Multimodal AI

Results : InvadeAI uses multimodal AI to generate rich, interactive advertisements that adapt to user context, making creatives more dynamic and improving customer engagement across digital channels.

Limitations : The system focuses on creative generation and interactive ad experiences only, without coupling ads to local demand forecasts, social trend lifecycles, or stock and inventory conditions for SMEs, leaving the execution loop incomplete.

2.5 Paper 5 : Smart Retail – Utilising Machine Learning for Demand Prediction, Price Strategy, and Inventory Management

Results : ML-based methods for demand prediction, dynamic pricing, and inventory optimisation improve profit margins, reduce stockouts, and make replenishment more efficient in smart retail operations.

Limitations : The framework is largely enterprise-oriented, assuming rich structured data and infrastructure, and does not integrate live social media trends, influencer activity, and creative campaign tools into a unified, SME-friendly end-to-end platform.

2.6 Summary of Research Gap

Across the reviewed literature, a consistent finding emerges: existing studies address individual capabilities — demand forecasting, trend analysis, recommendation, or creative generation — in isolation. No existing work proposes a unified, SME-friendly platform that integrates hyper-local trend intelligence, micro-influencer discovery, AI content generation, receipt-level inventory tracking, and proactive alert delivery within a single closed-loop system. This gap constitutes the primary academic and practical motivation for Advantage OS.

Paper	Contribution	Limitation / Gap
Tiwari & Prasad [1]	AI-based retail demand forecasting	No social trend or campaign integration; enterprise focus
Jamalpur et al. [2]	Social media trend & sentiment analysis for marketing	No inventory linkage; no operational end-to-end flow
Tomar et al. [3]	AI recommender for e-commerce personalisation	No hyper-local context, influencer matching, or inventory
Dunton et al. [4]	Multimodal AI for interactive ad generation	No linkage to demand signals or stock levels
Singhal et al. [5]	ML-based demand prediction & inventory management	Enterprise-oriented; no social media or creative module

Table 2.1: Comparative Summary of Reviewed Literature

Chapter 3

PROBLEM IDENTIFICATION

The problem addressed by Advantage OS can be articulated across several interconnected dimensions that collectively create a significant operational disadvantage for SMEs in the current digital economy:

- SMEs face difficulties in identifying and responding to rapidly changing local customer preferences due to the lack of affordable analytics and forecasting tools.
- Advantage OS is proposed as an AI-driven Growth-as-a-Service platform that converts social media trends into useful marketing and inventory decisions for SMEs.
- The system is designed for a hyper-local digital economy where neighborhood-level trends strongly influence customer engagement, product demand, and business performance.
- Advantage OS integrates trend detection, influencer discovery, automated campaign generation, and inventory monitoring into a single unified platform.
- Unlike existing SME tools that focus only on analytics or campaign creation, the proposed system provides a complete closed-loop workflow from social listening and campaign execution to performance tracking and stock management.

The cumulative effect of these gaps is that SMEs operate in a reactive mode: responding to trends after they have peaked, launching campaigns without adequate creative support, and restocking only after they have already lost sales. Advantage OS is designed to shift this dynamic from reactive to proactive, from fragmented to integrated, and from enterprise-exclusive to SME-accessible.

Chapter 4

OBJECTIVES

The major objectives of the proposed project are:

- Detect and quantify hyper-local trends: monitor social platforms (Reddit, YouTube, etc.) to identify locality-specific topics, convert them into structured neighbourhood- or PIN-level trends, and forecast each trend's growth, peak, and decay with time-series models. Present easy-to-read scores (trend strength, local relevance, urgency) so non-technical owners can quickly prioritise opportunities.
- Drive targeted campaigns and influencer selection: enable SMEs to generate on-brand images and captions with generative AI, estimate on-device CTR/engagement before publishing, and improve ROI by clustering influencers by engagement, audience size, niche, and locality while filtering suspicious/botted accounts.
- Close the loop with inventory-aware execution: capture receipts via OCR and use ML on sales velocity and stock levels to predict stockout risk for trending products; send proactive WhatsApp alerts for “trend high, stock low” conditions and provide a unified, modular dashboard that integrates trend intelligence, influencer discovery, creative generation, and inventory analytics.

Chapter 5

METHODOLOGY

Advantage OS follows a modular, phased development approach. Each core module — Trend Engine, Influencer Matchmaker, Creative Studio, and Inventory Edge — is designed, built, and validated independently before integration into the unified dashboard. The system architecture reflects a closed-loop design where social signal ingestion drives campaign creation, campaign performance feeds back into trend scoring, and trend scores connect to inventory alerting.

5.1 System Architecture

The overall architecture of Advantage OS is organised into six layers: Data Sources, Data Ingestion & Storage, AI & Analytics Engine, Application/Service Layer, User Interface, and Output & Integrations. Figure 5.1 illustrates this architecture.

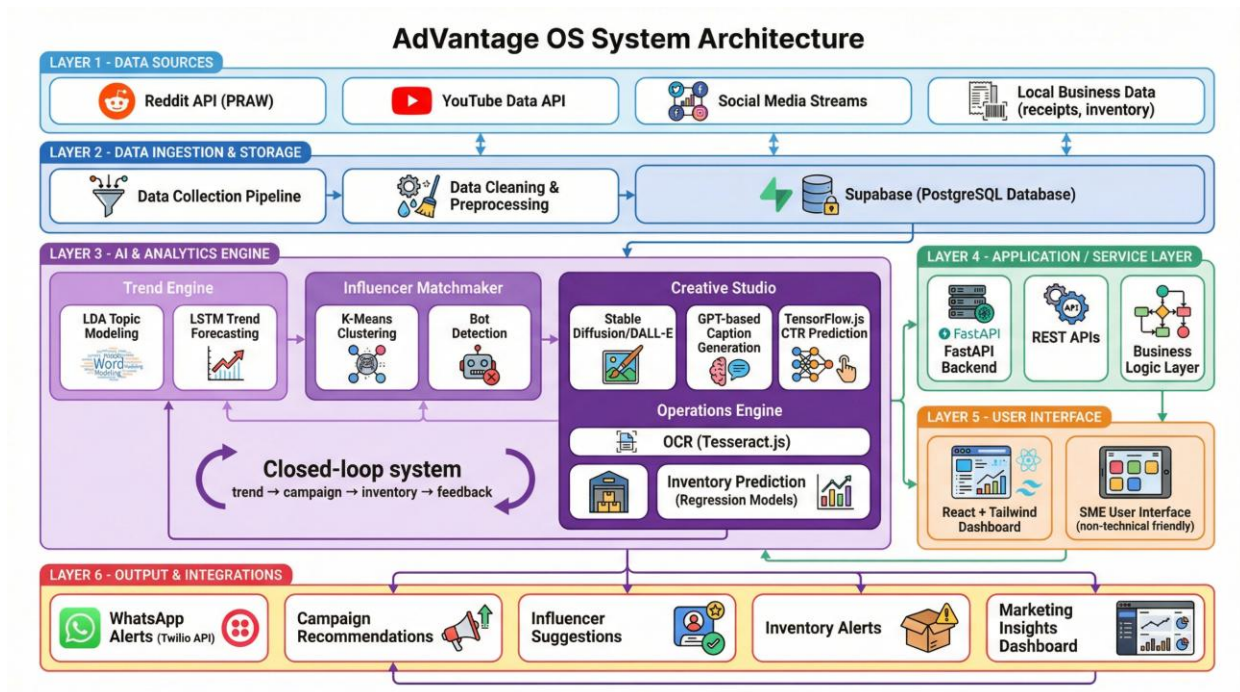


Figure 5.1: Advantage OS System Architecture Diagram

Layer 1 (Data Sources) includes Reddit API via PRAW, YouTube Data API, social media streams, and local business data such as receipts and inventory records. Layer 2 (Data Ingestion & Storage) encompasses the data collection pipeline, cleaning and preprocessing steps, and the Supabase

PostgreSQL database. Layer 3 (AI & Analytics Engine) contains the Trend Engine (LDA + LSTM), the Influencer Matchmaker (K-Means + Bot Detection), and the Creative Studio (Stable Diffusion, GPT captioning, TensorFlow.js CTR). Layer 4 (Application/Service Layer) includes the FastAPI backend, REST APIs, and business logic. Layer 5 (User Interface) delivers a React + Tailwind dashboard and an SME-friendly interface. Layer 6 (Output & Integrations) produces WhatsApp alerts, campaign recommendations, influencer suggestions, inventory alerts, and the marketing insights dashboard.

5.2 Module 1 – Data Ingestion

The Data Ingestion module is the entry point for all external information into AdVantage OS. It is responsible for collecting structured and unstructured data from public social media APIs, cleaning and normalising it, and persisting it in the Supabase database for downstream processing.

Relevant subreddits, YouTube channels, and keyword filters are first identified based on the SME niche being served (e.g., food, fashion, wellness in Bengaluru). The Reddit PRAW library and YouTube Data API are then used to periodically fetch posts, video metadata, comments, timestamps, and engagement signals. Data cleaning involves removing duplicates, stripping HTML and emoji characters, handling Unicode normalisation, and resolving language inconsistencies. All cleaned records are stored in PostgreSQL with schemas separating trends, posts, influencer profiles, campaign records, and inventory entries.

Background cron jobs are scheduled to run ingestion pipelines at configurable intervals (e.g., every 4–6 hours for Reddit, daily for YouTube) to keep the trend intelligence fresh without exceeding API rate limits.

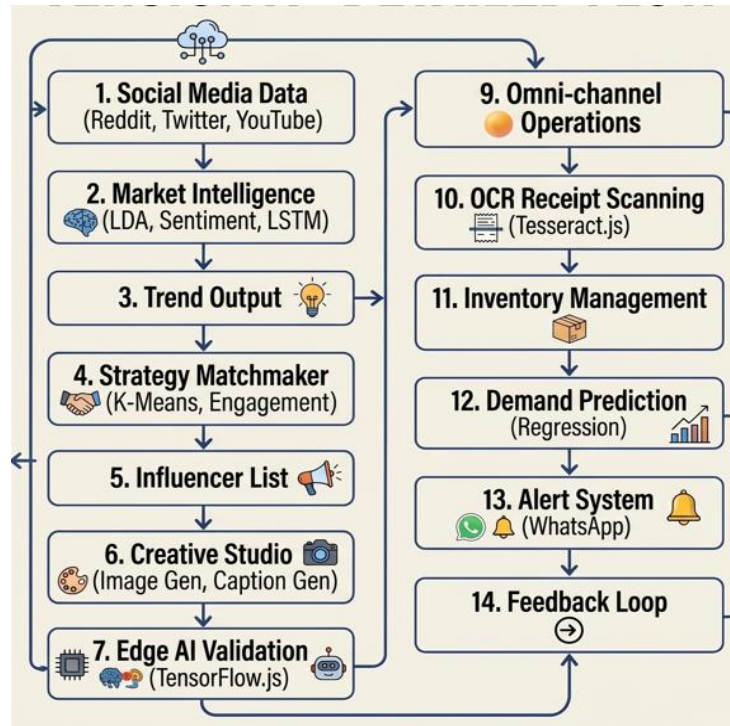


Figure 5.1: System Workflow Diagram

5.3 Module 2 – Campaign Creation and Launch (Trend Engine + Influencer Matchmaker + Creative Studio)

This module encompasses three interconnected components that together transform raw social data into a ready-to-launch marketing campaign.

Trend Engine: Text preprocessing steps (tokenisation, lowercasing, stop-word removal, and lemmatisation) are applied to social media text. LDA topic modelling is then applied to cluster posts into coherent topics representing potential local trends. For each identified topic, a time-series of activity (posts per day, engagement per day) is constructed and an LSTM model is trained to forecast trend strength over a short horizon. A composite 'window of opportunity' score is computed from predicted peak time, current growth rate, and locality relevance, and exposed as a colour-coded indicator in the dashboard.

Influencer Matchmaker: Candidate influencer profiles are collected from YouTube and Instagram sources, with features including follower count, engagement rate, content tags, and geographic location. K-Means clustering groups influencers by niche, audience size, and engagement pattern. Bot and fraud detection rules — including abnormal follower/following ratios, very high likes with

minimal genuine comments, and repetitive content patterns — filter out unreliable accounts. For each active trend, an influencer–trend similarity score is computed and a shortlist of micro-influencers is recommended to the user.

Creative Studio: A React + Tailwind CSS dashboard enables users to select a trend and product and request creative suggestions. A text-to-image model (Stable Diffusion or DALL-E via API) generates ad images from trend-aware prompts. A language model generates multiple caption options aligned with the trend tone and audience. TensorFlow.js runs a browser-side CTR estimation model that scores each creative before the user selects one for launch.

5.4 Module 3 – Retail Operations (Inventory Edge)

The Inventory Edge module bridges the gap between online marketing activity and offline stock readiness. SME users upload or photograph receipt images through the web interface. Tesseract.js performs in-browser OCR to extract key fields: item name, quantity, unit price, and transaction timestamp. Extracted items are mapped to internal product IDs using fuzzy matching to handle OCR errors, and stock-in and stock-out records are updated in near real-time.

The module maintains approximate current inventory levels for monitored SKUs and exposes this data to the Analytics & Feedback module for stockout prediction.

5.5 Module 4 – Analytics and Feedback

This module combines trend forecasts, active campaign status, and sales velocity to compute product-level future demand estimates. A regression or classification model is trained to predict whether a product is at risk of stockout within a configurable time window (e.g., 3–5 days) given current trend intensity and sales rate. When the predicted risk exceeds a threshold, the Twilio WhatsApp Business API is triggered to send an automated alert to the business owner containing the product name, estimated days to stockout, and the trend driving demand.

All alerts and owner responses (reordered, ignored, campaign paused) are logged and fed back into model retraining pipelines, enabling the system to improve its threshold calibration and prediction accuracy over time. This feedback loop is the key element that makes AdVantage OS a closed-loop system rather than a one-way analytics dashboard.

5.6 Expected Outcome

The expected outcome of Advantage OS Phase 1 is a functional, integrated prototype demonstrating the end-to-end flow from social trend ingestion through campaign generation to inventory alerting. The system is expected to enable local SMEs to identify meaningful local opportunities earlier than competitors, generate well-aligned marketing campaigns faster, and proactively manage inventory in anticipation of trend-driven demand spikes.

Technical deliverables include a modular full-stack web application, trained or integrated ML models for trend forecasting, influencer clustering, and stockout prediction, API integrations with Reddit, YouTube, Stable Diffusion/DALL-E, and Twilio WhatsApp, a React dashboard, and comprehensive project documentation. Academically, the project demonstrates a practical multimodal AI architecture combining NLP, time-series forecasting, computer vision, OCR, and generative AI in a single coherent retail use case.

5.7 Work Plan / Timeline

Phase	Task	Duration
Phase 1	Problem understanding, literature review, and requirement analysis	Week 1-2
Phase 2	System design, dataset planning, and architecture finalisation	Week 3-4
Phase 3	Social data ingestion module and database setup	Week 5-6
Phase 4	Trend analysis, forecasting, and influencer discovery modules	Week 7-9
Phase 5	Creative Studio and dashboard development	Week 10-11
Phase 6	OCR-based receipt processing, inventory prediction, and alert module	Week 12-13
Phase 7	Integration, testing, evaluation, and performance refinement	Week 14-15
Phase 8	Final documentation, presentation preparation, and project review	Week 16

Table 5.1: Project Work Plan and Timeline

Chapter 6

REQUIREMENT SPECIFICATION

6.1 Software Requirement Specification

Frontend Technologies:

- React.js – Building the dashboard UI and campaign workflows.
- Tailwind CSS – Responsive, component-based styling of charts, cards, and forms.
- JavaScript / TypeScript (ES6+) – Interactive components, API calls, and client-side state management.
- TensorFlow.js – Browser-side image classification and CTR estimation.
- Tesseract.js – In-browser OCR for receipt scanning.

Backend Technologies:

- Python + FastAPI / Flask – REST API implementation, ML orchestration, and business logic layer.
- Supabase (PostgreSQL + Auth) – Persistent storage of users, trends, influencer records, campaign data, and inventory, with built-in authentication and row-level security.
- REST APIs – Communication between frontend, ML services, and external providers.
- Cron Jobs / Background Workers – Periodic data ingestion, model updates, and batch computations via Supabase functions or server-side schedulers.

AI and Machine Learning Tools:

- scikit-learn – LDA topic modelling, K-Means clustering, and regression models for CTR and inventory prediction.
- TensorFlow / Keras – LSTM time-series forecasting for trend strength prediction.
- Transformers (Hugging Face) – Sentiment analysis and text embeddings for trend scoring and caption generation.
- Stable Diffusion / DALL-E (via Hugging Face or OpenAI APIs) – Trend-aware ad image generation.
- GPT-style Language Models – Campaign caption generation, influencer outreach messages, and copy variants.

Messaging and Integration:

- Twilio WhatsApp Business API – Automated stock-risk and campaign notifications to SME owners.
- Reddit API (PRAW) – Periodic social media data ingestion from relevant subreddits.

- YouTube Data API – Video metadata, comments, and engagement data ingestion.

Datasets:

- Public Reddit posts and YouTube video data (titles, descriptions, comments, engagement) fetched via official APIs for local trend detection and forecasting.
- Influencer profile and engagement data from YouTube Data API and Instagram creator profiles for the Influencer Matchmaker.
- M5 Forecasting – Walmart public retail sales time-series dataset for benchmarking LSTM demand forecasting models.
- A Million News Headlines corpus for prototyping and validating LDA topic modelling pipelines.

6.2 Hardware Requirement Specification

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or higher (8th generation or above)
RAM	Minimum 8 GB (16 GB recommended for ML model training)
Storage	Minimum 256 GB SSD (for OS, datasets, and model artifacts)
GPU (Optional)	NVIDIA GPU with CUDA support recommended for LSTM and Stable Diffusion training
Operating System	Ubuntu 20.04 LTS / Windows 10 or higher / macOS 12+
Browser	Google Chrome / Firefox (latest version, for TensorFlow.js compatibility)
Network	Stable broadband internet connection (for API access and cloud database)
Cloud/Hosting	Supabase (PostgreSQL cloud), Vercel or Railway for backend deployment

Table 6.1: Hardware Requirements

Chapter 7

CONCLUSION

Advantage OS addresses a well-defined and significant gap in the current landscape of digital marketing and retail operations tools available to small and medium enterprises. The platform proposes a unified, closed-loop Growth Operating System that begins with hyper-local social trend detection, proceeds through AI-assisted campaign creation and micro-influencer matching, and closes the loop with inventory tracking and proactive stockout alerting — all delivered through a single, non-technical-friendly dashboard.

Phase 1 of the project has established a solid foundation: the problem has been clearly identified across multiple interdependent dimensions, a comprehensive literature review has contextualised the work within existing research and validated the novelty of the integrated approach, the system architecture has been defined across six layers with four core modules, and the software and hardware requirements have been fully specified.

The proposed technical stack — React + Tailwind CSS on the frontend, Python FastAPI on the backend, Supabase for data persistence, and a suite of ML and generative AI tools including scikit-learn, TensorFlow/Keras, Hugging Face Transformers, Stable Diffusion, and Twilio — is both technically sound and academically defensible, drawing on widely validated tools used in production systems.

The significance of this project extends beyond its immediate deliverables. By demonstrating that NLP, time-series forecasting, generative AI, computer vision, and OCR can be combined within a single SME-focused operational system, AdVantage OS contributes to the broader academic discourse on AI democratisation — the principle that advanced machine learning capabilities should be accessible to small businesses, not just enterprise organisations with large technical teams.

Upcoming phases of the project will focus on implementing and validating each module independently, integrating them into the unified dashboard, evaluating system performance against defined metrics (trend forecasting accuracy, influencer recommendation quality, OCR accuracy, stockout alert lead time), and conducting user testing with SME-representative participants. The timeline outlined in Chapter 5 provides a structured roadmap for achieving these milestones within the academic year.

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